

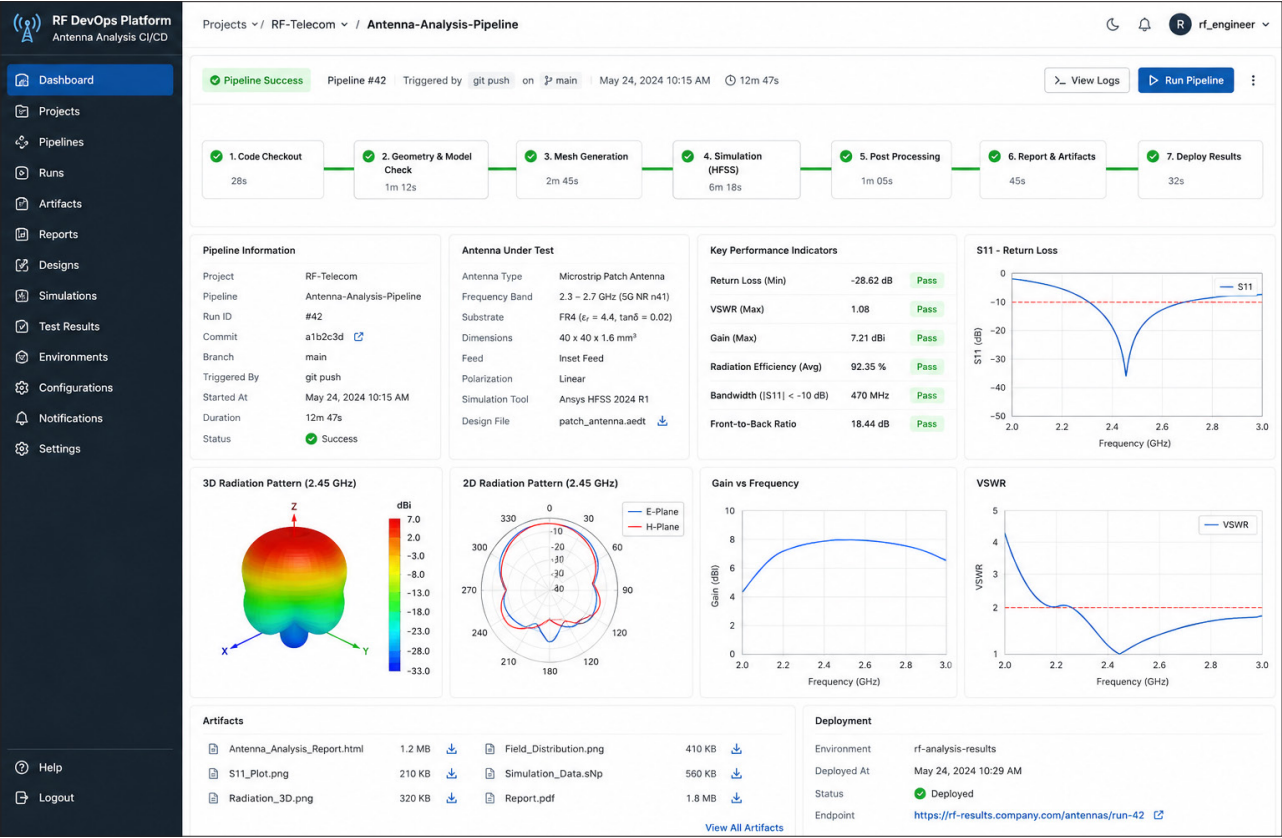


Figure 3: DevOps pipeline and dashboard for RF engineering and telecom

Table 2: Popular open source toolkits for DevOps in RF and SDR

Tool/Library	URL	SDR/RF application	Use case
GNU Radio	gnuradio.org	Core engine for defining radio waveforms and modulation	Radio frequency intelligence, Digital signal processing (DSP), Wireless analytics
GitLab CI/CD	gitlab.com	Managing multi-stage builds and triggering remote tests	Orchestration and Pipelines
Labgrid	labgrid.org	Automating interaction with physical SDR hardware	Hardware-in-the-Loop (HIL)
Docker	docker.com	Standardising DSP build environments (UHD, libraries)	Containerisation
GPS-SDR-SIM	github.com/osqzss/gps-sdr-sim	Creating synthetic GNSS signals for receiver testing	Signal simulation
Buildroot	buildroot.org	Creating lean, reproducible images for embedded SDRs	Firmware deployment
Ansible	ansible.com	Automating remote SDR node updates and configuration	Configuration management

defence and military applications, the auto-detection of suspicious or unknown flying objects can lead to swift decision making. Researchers can make use of GNU Radio, Python libraries and other similar platforms

for real-time air signal analysis, predictions, and decision making. **END** 

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